AN ENGINEERING DOCUMENTARY

Subject: COMPREHENSIVE TECHNICAL REPORT: FIELD EFFECT TRANSISTORS (FETs)

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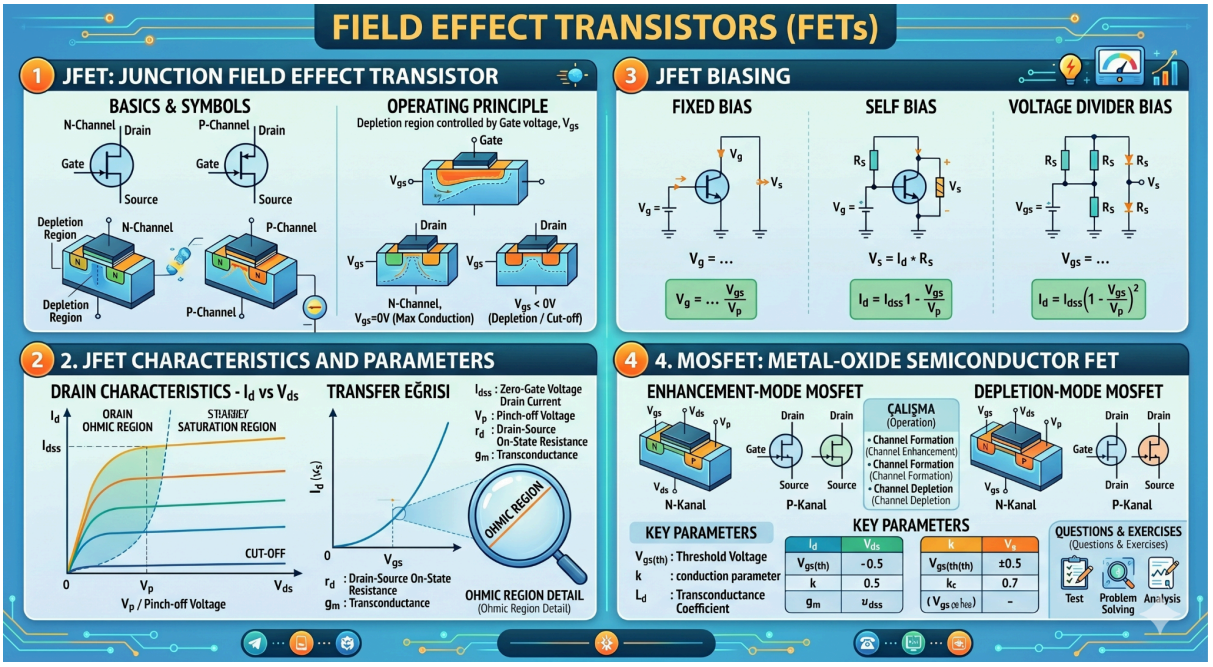

Field Effect Transistors (FETs) are voltage-controlled semiconductor devices used to regulate current flow between terminals. They offer high input impedance, low noise, and are widely used in both analog and digital circuits. MOSFETs, in particular, are the foundation of modern electronics such as microprocessors and power systems.

1. Introduction to Field Effect Transistors (FETs)

The Field Effect Transistor (FET) is a three-terminal semiconductor device that controls the electrical current flowing between two terminals (Source and Drain) by applying an electric field to a third terminal (Gate).

Unlike the Bipolar Junction Transistor (BJT), which is a current-controlled device (base current controls collector current), the FET is fundamentally a voltage-controlled device (gate voltage controls drain current).

Because its operation relies on only one type of charge carrier—either electrons (n-channel) or holes (p-channel)—it is classified as a unipolar device.

1.1 Fundamental Operating Principle

In a FET, the gate voltage modulates the conductivity of a semiconductor channel. When a voltage is applied to the gate, it alters the width of a depletion region within the channel, thereby increasing or decreasing the resistance between the drain and source terminals.

This voltage-controlled resistance is the core mechanism enabling amplification and switching.

1.2 Key Advantages of FETs Over BJTs

Parameter	FET Advantage	Technical Justification
Input Impedance	Extremely High (MΩ to TΩ)	Gate draws negligible current
Noise Figure	Low Noise	Less shot and recombination noise
Thermal Stability	High	Negative temperature coefficient
Manufacturing Density	Very High	Scales to nanometers
Offset Voltage	Near Zero	No base-emitter requirement

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2. The Junction Field Effect Transistor (JFET)

The JFET is the earliest type of FET. It consists of a semiconductor channel with a gate region of opposite doping.

2.1 Structure and Operation

- n-channel: electrons carry current
- p-channel: holes carry current

Reverse bias on the gate widens the depletion region and reduces current.

2.2 Characteristics

Regions of Operation

1. Ohmic Region
    - Acts like a resistor
    - Low V_DS
2. Saturation Region
    - Acts like current source
    - Used for amplification
3. Breakdown Region
    - Device damage risk

Shockley Equation

$ID = IDSS * (1 - VGS / VGS(off))^2$

Where:

- IDSS = max drain current
- VGS(off) = cutoff voltage

2.3 Biasing Methods

Method	Stability	Use
Self Bias	Medium	Simple circuits
Voltage Divider	High	Precision circuits
Current Source	Very High	IC design

Self-bias formula:

$VGS = -ID * RS$

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3. Ohmic Region (Voltage Variable Resistor)

JFET behaves like a resistor when VDS is small.

Approximate resistance:

$rds \approx VP / (2 * IDSS * (VGS - VP))$

Applications

- AGC systems

- Oscillators
- Analog control circuits

Limitation: signal must be small to avoid distortion.

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4. MOSFET (Metal-Oxide Semiconductor FET)

MOSFET has an insulated gate using SiO_2.

4.1 Key Properties

- Very high input resistance
- Gate capacitance present

4.2 Types

Feature	Depletion MOSFET	Enhancement MOSFET
Channel	Exists	Induced
Default State	ON	OFF
Usage	Analog	Digital

4.3 MOSFET Equation (Saturation)

$ID = kn * (VGS - Vth)^2$

Where:

- kn = device constant
- Vth = threshold voltage

4.4 Biasing Methods

Method	Stability
Drain Feedback	Good
Voltage Divider	Very Good

Current Mirror	Excellent

4.5 CMOS Technology

CMOS uses both nMOS and pMOS.

- Zero static power
- Used in all processors

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5. Practical Considerations

5.1 ESD Sensitivity

MOSFETs are highly sensitive to static electricity.

5.2 Thermal Behavior

More stable than BJTs.

5.3 Reliability

No second breakdown issue like BJTs.

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6. Comparison Table

Property	BJT	JFET	MOSFET
Control	Current	Voltage	Voltage
Input Impedance	Low	High	Very High
Noise	Medium	Low	Low
Speed	High	Medium	Very High
Usage	Analog	RF	Digital/Power

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7. Conclusion

FETs represent a major shift from current-controlled to voltage-controlled electronics.

- JFET: low-noise analog applications
- MOSFET: dominant in modern electronics

MOSFET technology powers:

- microprocessors
- power electronics
- digital systems

Future technologies like FinFET and GAA FET continue this evolution.